



**MASTERMIND
INDIA**
Spirit of India Special

BREATHE EASY

**Dr Aarti
KINIKAR, 46**

INNOVATION

**Indigenous
nasal bubble**

COST

Rs 250

WHAT IT DOES

**Delivers oxygen to
children with respira-
tory ailments and
steadies the air sacs**

At the height of the swine flu outbreak in Pune last year, Sasson Hospital only had four ventilators and was faced with an influx of 900 children suffering from respiratory ailments. Quick thinking by Dr Aarti Kinikar, who devised an indigenous nasal bubble to deliver oxygen, saved hundreds of lives. Kinikar, who heads the swine flu and thalassemia units at the government hospital's paediatrics ICU, used local devices such as a saline bottle, an iv fluid transfusion and a paediatric nasal cannula to deliver oxygen to the children's lungs. "Usually, such a device is used only for newborns but we used it for children up to 12 years old. I took a risk. There was no choice, you have to save lives—you cannot have patients dying," she says.

The innovation is based on a device called a nasal bubble Continuous Positive Airway Pressure, which costs Rs 5 lakh. Kinikar was able to create a similar device with local materials that cost only Rs 250. One end of the nasal cannula, a small tube was attached to the patient's nose and the other end to

a central oxygen flow meter. "Apart from giving Tamiflu, the other main treatment in swine flu is to give oxygen. The air sacs collapse during swine flu so it's important to keep them open. When the patient exhales, bubbles are formed in the saline, which ensures the air sacs do not collapse," she says.

Between August 2009 and March 2010, 21 swine flu positive patients and 310 with suspected symptoms were given the treatment. All swine flu patients were transferred to Sasson as private hospitals were unwilling to deal with them. "There would have been several mortalities without this device," she says. Moreover, Kinikar says the local device was better than a ventilator, which is an invasive procedure where the risk of secondary infection goes up. But she says the device should only be used if it shows signs of working within six hours.

**THE INDIGENOUS
NASAL BUBBLE
DEVICE HAS TURNED
OUT TO BE A VIABLE
ALTERNATIVE TO
A VENTILATOR.**

The procedure is now being routinely used in the hospital and has been adopted by seven small hospitals and nursing homes in Maharashtra as she conducted workshops for doctors on how to use it. Apart from this, she has also developed an ICU for thalassemia and has helped upgrade a neo natal ICU and newborn care. "Innovation is crucial for medicine. That is how medicine has progressed. Innovation lowers the cost of healthcare, thereby helping the patient. I am interested in treatments that will help the patient directly at the bedside," says Kinikar. **by Nishika Patel**



YUSUF KHAN

**DR KINIKAR
SHOWS OFF
THE DEVICE**